

Quality assessment, prognostic factors, and biomarkers for brain tumor analysis: a comprehensive systematic review

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ABSTRACT

The brain tumors possess different causative factors and properties, making their diagnosis and treatment difficult. Growth of these cancers usually leads to compression of the adjacent nerves and obstruction of the flow of cerebrospinal fluid, thus leading to increase in intracranial pressure. This affects the working of brain in many ways; thus, the difficulty involved in its treatment. With the improvements in technology in neuroimaging, including Diffusion Tensor Imaging (DTI), Positron Emission Tomography (PET), and multiparametric Magnetic Resonance Imaging (mpMRI), the diagnosis process has become easy. The effectiveness of any form of therapy in such patients depends primarily on their prognosis. While it is a common practice that physicians determine the prognosis of the disease by considering the age of the patient, histological grade of the tumor, and resection status, now this method has become more comprehensive by adding molecular signature and genetic analyses to the list of criteria. Next-generation sequencing (NGS) allows a reliable molecular classification. It increases the level of risk stratification, facilitating the application of therapies tailored to individual patients. Thus, molecular oncology has greatly changed our views on brain tumors' pathology and prognosis while neoadjuvant treatments aim at increasing the survival rate. On the other hand, radiogenomics is a field of study that combines non-invasive imaging phenotypes and genomic information in order to find unique molecular signatures of tumors without collecting samples from tumors. Molecular biomarkers are absolutely essential in the diagnosis of cancer, treatment monitoring, and recurrence of cancer. Advances in liquid biopsy technology, particularly the methods for circulating tumor DNA (ctDNA) and Extracellular Vesicle (EV) based analysis, have enabled the possibility of non-invasive monitoring of the progression of the tumors over time. This review highlights key studies and important scientific works about imaging technologies, biomarkers, and prognostic factors of malignant brain tumors.

1. Introduction

The brain is the master gland of the nervous system, which has an essential function in controlling complicated biological processes required for life, cognition, and homeostasis. The occurrence of any kind of lesion within this organ, particularly cancer, leads to severe neurobiological problems. Because of their biologically invasive nature, infiltrative capacity, and unpredictable course of disease, brain tumors remain an important research topic in modern biomedicine. The lack of control and regulation in the growth of the neoplastic cells becomes another dimension of the choice of therapy issue. In the last decade, there has been much advancement in the field of brain tumor diagnostics and treatment due to direct breakthroughs made in the field of molecular biology and imaging technologies [1]. Molecular biology

techniques have enabled us to understand the underlying mechanisms of these tumors. Moreover, imaging techniques have enabled the early diagnosis of these tumors along with the visualization at a high resolution that helps formulate treatment strategy for the patient. The implementation of AI, in particular, the approach of deep learning, represents the main technological advancement that became feasible due to the development of radiation oncology and diagnostic radiology. The potential for clinical application of automatic lesion detection via networks is very high. In addition to the basic application of these approaches to oncology treatment, there are also practical applications of AI to other biomedical disciplines, such as immunology and allergology, when the predicting capacities of computational algorithms are optimized. Currently, the MRI remains the gold-standard imaging technique for brain tumors due to superior soft-tissue contrast and multi-planar

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acquisitions. The imaging engine plays a crucial role in helping obtain important anatomical data related to the location, volumes, structure, and spatial relationship of the tumor with respect to the surrounding brain tissue. Despite the high level of effectiveness of the conventional MRI approach, there are still some difficulties when classifying specific molecular subtypes and histopathological features [2]. The problems in making a diagnosis arise due to different views of observers, similarity of radiological appearance of various kinds of tumors, and intratumor heterogeneity. In order to overcome the issue, it is important to develop computing systems which will link radiological data with molecular profiling. Due to the complexity of the molecules associated with brain tumor biology and phenotype, there is a need for advanced analysis techniques in order to obtain correct differential diagnoses. Although conventional diagnostic techniques have proved to be efficient, they lack sufficient complexity with regard to the information available through the new radiology imaging techniques. This limitation underscores the need for employing automatic computational algorithms together with the multimodal technique. However, new developments that have been achieved within the sphere of deep learning have enabled great improvement of the abilities of analysis of the neuroimaging database. Through the use of MRI databases along with corresponding histopathological and follow-up information, these neural networks become capable of understanding the highly complicated features that are associated with different types of tumors. Feature space with high dimensionality that includes detailed information about the morphology, texture, signal intensity, and spatial distribution of the tumors may be easily captured using deep learning techniques with the use of Convolutional Neural Networks [3]. It removes the subjectivity of human intervention in the process of tumor classification thereby making the diagnostic procedure objective and consistent. The feasibility of this approach is gradually getting proved with actual facts. For instance, a recent study revealed that a deep learning algorithm was successfully used for diagnosing brain tumors based on magnetic resonance data with a success rate of 99.5%, which was much higher than traditional machine learning algorithms and manual diagnosis [2]. The above-mentioned performance indicates the potential of AI to enhance the techniques of neuro-oncological imaging. Nonetheless, it should be noted that in spite of all the successes achieved, specialists emphasize the importance of additional validation in clinical trials involving more diverse cohorts of patients. The key technical problem that remains unsolved today is the detection of micro-lesions and disease progress at the initial stages of the process. Thus, although the current indicators show excellent performance of deep learning algorithms in terms of diagnostics, they also reveal some weaknesses in the field.

Detection of microlesions and early stages of tumors being complicated from the technological standpoint requires additional investigation for the improvement of the diagnosis and prognosis of the patients. Within neuro-oncology, the finding of biomarkers is especially important since it allows defining and treating the disease [4]. Biomarkers make it possible to differentiate various cancer tumor types and to estimate their prognosis. Epigenetic modifications of the promoter of O6-methylguanine-DNA methyltransferase (MGMT) gene form one of the main molecular biomarkers used in glioma treatment. In particular, the gene produces a nuclear protein which helps repair O6-alkylated lesions at guanine DNA bases. The suppression of the MGMT gene by methylation of its promoter causes the inability of the specific repair mechanism, hence making the cancer cells more sensitive to the action of alkylation-based therapies such as temozolomide. Therefore, the assessment of MGMT promoter methylation is now an integral component of the clinical neuro-oncology pipeline. The patients who have positive MGMT promoter methylation are more sensitive to alkylation-based treatment and survive for a longer period than those that lack MGMT promoter methylation. Clinical significance of MGMT promoter methylation has been well validated through extensive studies conducted to test its predictive and prognostic importance. Clinical analysis of patients with glioblastoma validated the prognostic value of the

methylation status of O6-methylguanine-DNA methyltransferase (MGMT) promoter, along with other clinicopathological factors [5]. The clinical study verified the importance of MGMT promoter methylation as a molecular marker for personalized cancer therapy in neuro-oncology. At the same time, the study emphasized the importance of other molecular markers like mismatch repair system activity and different S-phase cell cycle proliferation biomarkers. The combined effect of these biological factors shows that the prognosis modeling of glioblastoma is always multidimensional [5]. It is because of this that the incorporation of molecular biomarkers and traditional clinical data can be considered an exceedingly logical approach for future clinical decision-making models. Such a model requires specific research on the biological variables in question to maximize their use in clinical decision-making systems.

Many clinical studies have shown that methylation of the MGMT gene promoter is a highly reliable marker for predicting and prognosis in patients suffering from glioblastoma, especially concerning resistance of cells to alkylating chemotherapy like temozolomide [6]. Those patients who have proven to be undergoing the epigenetic silencing of the MGMT promoter are more responsive and have increased chances of prolonged survival than those patients without methylation. In addition, the methylation of the MGMT gene promoter is associated with an increase in therapeutic tolerance, making this epigenetic marker very useful. This process is an extremely important milestone in the medical use of molecular markers for the modification and improvement of the treatment strategy for this type of cancer. Molecular mechanisms that underlie the correlation and its prognostic significance are thoroughly detailed within the established biomedical literature [5]. This review article emphasizes that the combination of epigenetic biomarkers like methylation of MGMT promoter with clinical variables helps in determining the prognosis of the patients along with personalized approaches for treatment. In addition to MGMT, there are other types of epigenetic changes in neuro-oncology, which include DNA repair proteins that can serve as valuable prognostic markers for tumorigenesis and genomic stability. These proteins work through recognizing and repairing any defective structure within DNA, and thus act as synergetic systems that limit mutation and ensure the stability of the genome, which are key to the process of oncogenesis. Thus, knowledge of how these DNA repair pathways operate is essential for developing precision medicine approaches for the treatment of intracranial tumors. One such important DNA repair protein is the O6-methylguanine-DNA methyltransferase (MGMT). MGMT's demethylation is directly targeted at the O6 position of guanine, which represents one of the main sites where alkylations take place in DNA. In turn, MGMT functions as an antidote to alkylating agents used in therapy because it eliminates their cellular toxic effect.

Thus, the epigenetic methylation of the MGMT gene acquires high clinical importance since it can regulate the effectiveness of alkylating chemotherapy, along with the overall patient prognosis [7]. The combined results reveal that it is essential to incorporate molecular biomarkers into the workflow of neuro-oncology practice. Through the integration of relevant genomic, epigenomic, and clinical-pathological features, the principles of precision oncology can be better utilized to enhance patient care. Hence, the ultimate aim of this review paper is to equip neuroimaging practitioners and neuro-oncologists with an analysis of pertinent genetic and molecular biomarkers, particularly with respect to their biology and relevance in current brain tumor treatments. Personalized medicine constitutes a revolutionary paradigm that has emerged in modern clinical practice, whereby unique genomics and proteomics constitute fundamental elements in the diagnosis, treatment, and prevention of various diseases. In neuro-oncology, the personalized approach has become extremely important due to high molecular diversity typical of brain tumors. During the last ten years, a significant number of experimental studies have shown that specific genetic changes in certain cells are responsible for oncogenesis. This type of genetic changes includes a wide range of abnormalities on the molecular level, ranging from single nucleotide mutations to chromosome

abnormalities and gene fusion. Genomic analysis done in combination with sophisticated computational methods reveals valuable information about cancer biology [8]. Such utility is mostly due to the ability of bioinformatics workflow and machine learning framework to reveal replicable mutation profiles which are hard to determine otherwise. Additionally, these integrative strategies enable advanced tumor classification models and help to quickly identify new targets for treatment. Computation tools are also very important in the analysis of huge amounts of information produced by various sequencing databases to map the connections between the individual genetic variations and particular clinical outcomes. This enables the identification of patient-specific risks and the use of therapies tailored to the molecular subgroup by targeting the identified therapy targets. Taken together, all these achievements suggest a major change in paradigms from discovery-oriented approach towards a data-oriented one when dealing with neuro-oncology. All these achievements show that the integration of computational biology and molecular genetics changes clinical neuro-oncology. This suggests that future developments in this field will play a key role in using sequencing data in clinical practice.

1.1. Classifications

Age-related stratification is one of the essential preconditions for adequate treatment planning and accurate diagnostics within neuro-oncology. The histological makeup and biological properties of intracranial tumors within an adult population vary significantly from that which occurs in children's groups in terms of pathogenic features, anatomical location, and prognosis. Age-specific approaches to treatment ensure optimal efficiency while minimizing possible side effects. However, pediatric brain tumors differ greatly from their adult counterparts in terms of their biological behavior in a number of different respects [9]. It is important to remember that pediatric brain tumors are unique biological entities and not simply smaller versions of adult tumors; therefore, adult treatment methods cannot be extrapolated to pediatrics. The common primary brain tumors in children include medulloblastoma, ependymoma, craniopharyngioma, and pilocytic astrocytoma, where each of the mentioned pathological types is characterized by certain unique features on histological and molecular level [10]. From geometrical point of view, childhood brain tumors tend to occur in specific compartments of the brain, in particular in the posterior fossa, and they develop as a result of different aberrant development signaling pathways including Sonic Hedgehog pathway for certain medulloblastoma types. The problem arising with pediatric therapy is that developing neural tissue is highly radiosensitive and chemotherapy-sensitive, which causes long-lasting complications.

In contrast, intracranial cancers in adults have unique epidemiologic characteristics and molecular markers. The major types of brain cancers seen in adults include glioblastoma, meningioma, pituitary neuroendocrine tumors, and anaplastic astrocytoma [11]. Adult brain oncogenesis involves distinct molecular changes in certain genes like the Epidermal Growth Factor Receptor (EGFR), Phosphatase and Tensin Homolog (PTEN), and Isocitrate Dehydrogenase 1 and 2 (IDH1/2), all of which are crucial for tumor development and treatment. In addition, management of adults requires careful consideration of their associated comorbidities, baseline function, and systemic tolerance to treatments. Such comprehensive genomic analyses have likewise contributed to improvements within current age-based classification systems. The molecular analysis of such tumors has identified well-defined genetic causes of cancer formation in various types of neoplasms, which include IDH mutant glioblastoma and IDH wild-type glioblastoma [12]. The integration of such molecular features within existing classification systems improves overall diagnostic and prognostic precision.

In fact, the availability of extensive genomic information has been instrumental in promoting the incorporation of molecular pathology

into classification systems that were traditionally based on morphological histology, for classifying intracranial tumors. This, in turn, has greatly increased the precision and biological specificity of brain tumor classification, allowing for the implementation of highly specific and personalized therapy regimes depending on the genetic makeup of each tumor. An outline of brain tumor classification based largely on microstructure and cell morphology is illustrated in Fig. 1.

1.1.1. The WHO classifications

The incorporation of molecular criteria in addition to traditional histopathology in the World Health Organization classification scheme of central nervous system cancers represents a clear change of paradigm in neuro-oncological diagnosis. This new diagnostic strategy represents an evolution from purely morphologically based classification schemes to classification systems driven by biology, which better represent intrinsic features of tumors [13]. This introduction of the guideline is an important milestone in that it marked the inclusion of the molecular biomarkers in standard diagnostic algorithms, and the introduction of major structural changes to the classification of diffuse gliomas, medulloblastomas, and other primitive embryonal tumors. The key advancement in the context of this molecular classification is the recognition of diffuse midline glioma, H3 K27M-mutant, which underlines the importance of certain changes in histones for prognosis. Simultaneously, the classification of glioblastomas according to Isocitrate Dehydrogenase mutation has distinguished two phenotypes: IDH-mutant vs. IDH-wildtype.

Clinical use of the integrated molecular approach has greatly improved the accuracy of diagnosis while reducing the possibility of inter-observer variability in interpretation. By going beyond simply using descriptive histopathology to include unique genetic and epigenetic characteristics, the classification system used by the World Health Organization better represents tumor biology. As a result, this multidimensional approach plays a key role in the clinical application of treatment strategy selection and risk stratification. This is because incorporating the unique molecular characteristics makes it possible to tailor therapies for particular tumors [13]. The transition into molecular stratification is an important breakthrough in the field of precision medicine and has enabled healthcare professionals to develop specific treatment regimens purely on the basis of the genetic makeup of the disease in question. All these guidelines together constitute an ongoing development towards a multi-dimensional diagnosis that encompasses the morphology, molecular biology, and phenotypes of cells. A comparative summary of the key features of this classification of brain tumors is given in Table 1.

The classification criteria by the World Health Organization in 2016 of the tumors of the central nervous system is undoubtedly an innovative development in the sphere of neuro-oncology. The incorporation of the molecular markers into the assessment along with histopathological analysis makes the diagnostic process more efficient. The introduction of distinct genetic and epigenetic changes into the new diagnostic criteria increases the accuracy of prognosis and contributes to personalized therapy [14]. The classification of tumors according to such intrinsic molecular characteristics improves the process of patient risk stratification and provides an optimal link between the diagnosis classification and the biological nature of the disease. The systematic presentation of distinct types of central nervous system tumors, including their particular characteristics and primary prognostic factors, is shown in Table 2.

Classification system of central nervous system tumors developed by World Health Organization was a significant milestone in clinical neuro-oncology as it created the first ever taxonomy that helped incorporate molecular criteria in the traditional histological criteria of defining unique intracranial tumors. This was done through incorporation of individual molecular aberrations in the tumor nomenclature, thus providing more accurate diagnosis and taking into account the

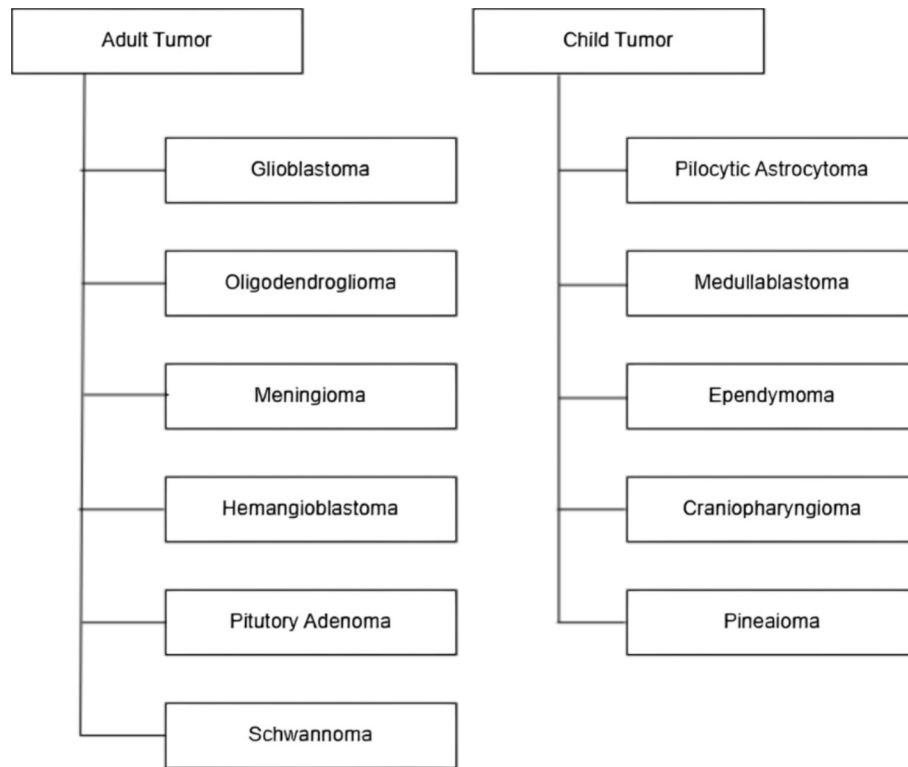


Fig. 1. Classification of brain tumors.

Table 1

Overview of WHO classifications for CNS tumors.

GradeType	WHO Grade I	WHO Grade II	WHO Grade III	WHO Grade IV
		Low Grade	High Grade	
Astrocytoma	Pilocytic Astrocytoma	Low-Grade Astrocytoma	Anaplastic Astrocytoma	Glioblastoma
Oligodendroglioma	–	Low-Grade Oligodendroglioma	Anaplastic Oligodendroglioma	–
Oligo-Astrocytoma	–	Low-Grade Oligo-Astrocytoma	Anaplastic Oligodendroglioma	–

Table 2

WHO comprehensive classification of Central Nervous System (CNS) tumors: subtypes, characteristics, and prognostic factors.

Type of CNS Cancer	Subtypes and Other Parameters	Classification Criteria	Treatment Options	Prognostic Factors
Gliomas	Astrocytoma (Grade I-IV), Oligodendroglioma (Grade II-III), Glioblastoma (Grade IV), Diffuse midline glioma, H3 K27M-mutant, Molecular markers (IDH mutation, 1p/19q co-deletion)	Histological features, Molecular markers	Surgery, Radiation Therapy, Chemotherapy, Targeted Therapy	MGMT Promoter Methylation Status, IDH Mutation, 1p/19q Co Deletion
Meningiomas	Grade I, II, III (based on histopathological features), WHO grade I meningothelial meningioma, WHO grade II atypical meningioma, WHO grade III anaplastic meningioma	Histopathological characteristics	Surgery, Radiation Therapy, Observation	WHO Grade, Extent of Resection
Pituitary tumors	Non-functioning pituitary adenomas, Prolactin-secreting adenomas, Growth hormone-secreting adenomas, ACTH-secreting adenomas, Thyrotropin-secreting adenomas, Gonadotropin-secreting adenomas	Morphological features	Surgery, Medication (Prolactin Inhibitors, Somatostatin Analogs)	Hormonal Status, Tumor Size
Medulloblastomas and other embryonal tumors	Sonic hedgehog (SHH) subgroup, Wnt subgroup, Group 3, Group 4	Molecular subgroups	Surgery, Radiation Therapy, Chemotherapy	Molecular Subgroup, Extent of Resection
Astrocytomas	Pilocytic astrocytoma (Grade I), Diffuse astrocytoma (Grade II), Anaplastic astrocytoma (Grade III), Glioblastoma (Grade IV), Molecular markers (IDH mutation, MGMT promoter methylation)	Grade (I-IV), Molecular markers	Surgery, Radiation Therapy, Chemotherapy, Targeted Therapy	IDH Mutation, MGMT Promoter Methylation Status
Oligodendrogliomas	Grade II oligodendroglioma, Grade III anaplastic oligodendroglioma	Grade (II-III), Molecular markers	Surgery, Radiation Therapy, Chemotherapy	1p/19q Co-Deletion, Molecular Subtype
Ependymomas	Grade I myxopapillary ependymoma, Grade II ependymoma, Grade III anaplastic ependymoma	Histological features, Molecular markers	Surgery, Radiation Therapy, Chemotherapy	Age, Tumor Location
Choroid plexus tumors	Choroid plexus papilloma (Grade I), Choroid plexus carcinoma (Grade III)	Histopathological characteristics	Surgery, Radiation Therapy	Histopathological Features

tremendous biological diversity that is present within the same morphologically similar tumors. Nevertheless, despite these essential innovations, there have been some limitations with regard to diagnostics. Histopathological assessment was still seen as the main factor in determining the tumor grade and was widely used in risk stratification at the start [12]. As a result, while molecular biomarkers have been more frequently used for classifying the tumors in general, there was no necessity to use them in all cases.

Therefore, the classification process was still very dependent on the cellular morphology, which is inherently prone to interobserver interpretation variability. Despite the fact that Isocitrate Dehydrogenase mutations and 1p/19q co-deletion testing were encouraged as a means of facilitating stratification of diffuse gliomas, there was no obligation to apply molecular testing in the case of defining many other tumors of the central nervous system. As a result of the absence of any obligation to apply the molecular tests, it became difficult to implement nationwide uniformity between the histological and molecular classification systems since a number of tumors were classified only according to their microscopic features and not their genome. This diagnostic inconsistency made it absolutely necessary to make certain systemic improvements within the neuro-oncological classifications.

The study through observation brought out the drawbacks of the WHO Classification System of 2016 as being excessively dependent on complex molecular techniques that were not globally available [15]. During the management of diffuse gliomas and embryonic tumors, there was no facility for molecular diagnostics available. This information illustrated a significant gap between the potential benefits of molecular classification and the reality of applying this form of taxonomy to pathology departments, especially when applied in resource-limited clinical environments. The dependency on basic histopathology, the non-mandatory use of molecular markers, the presence of variations in diagnostics, and an unclear classification of prognostics all served to illustrate the need for a standardized diagnostic approach.

Significantly, the operational weaknesses observed in the 2016 model became the catalysts for further development in the field of neuro-oncological classification. The necessity of conducting certain tests was reiterated and reinforced in new diagnostic protocols, as well as the requirement for employing a hierarchical report that combines structural pathology with molecular knowledge into one diagnostic entry. As such, increased diagnostic accuracy, structural organization, and significance in terms of prognosis has been accomplished through a systematic reclassification of tumors according to somatic genetic mutations as well as a close association between histologic grading and biological behavior. Such a multi-faceted approach is clinically imperative, as the development of the current classification system through tumor biology maximizes clinical outcomes.

WHO CNS5: World Health Organization Classification of Tumors of the Central Nervous System, Fifth Edition introduces a groundbreaking paradigm shift in the field of neuro-oncology classification systems. Whereas earlier editions of these classification criteria provided information on the molecular aspects and pathological features independently of one another, the current system necessitates that both the histopathological characteristics and molecular features, along with all other intersecting clinical information, be merged into a single comprehensive diagnosis. The above mentioned structurally integrated system matches exactly the international standards for oncological diagnosis since there is a scientific consensus that the structural morphology alone is inadequate for the adequate representation of intrinsic tumor pathobiology.

The main achievement in the new taxonomy is the formal establishment of tumor entities that have been proven to be genetically determined. Based on the unified criteria of these guidelines, tumors of the central nervous system are increasingly classified according to the distinct types and subtypes defined on the basis of particular somatic genetic changes. Thus, in the case of diffuse astrocytomas with Isocitrate Dehydrogenase (IDH-wild type i.e., without any IDH mutation), it is

possible to diagnose definitively the central nervous system WHO grade 4 tumor only by the detection of certain molecular markers, such as Telomerase Reverse Transcriptase (TERT) promoter mutation, Epidermal Growth Factor Receptor (EGFR) amplification, or simultaneous gain of chromosome 7 and deletion of chromosome 10. In the case of diffuse astrocytomas with IDH mutations, it has been established that the homozygous deletion of the Cyclin-Dependent Kinase Inhibitor 2 A/B (CDKN2A/B) locus is a key, independent molecular predictor of CNS WHO grade 4.

Clinical importance of this multivariate approach comes precisely from the fact that it allows reducing subjectivity that had traditionally been connected with histological grades [13]. Using an objective system of diagnosis based on genetic and epigenetic abnormalities, this new classification scheme allows increasing evaluation accuracy and improving correlation of diagnostic results and patient survival. In addition to changes in molecular criteria, the modified taxonomy also features significant structural and terminology changes. Terms of older classifications that relied solely on histological criteria and had no biological or clinical significance were outdated or assimilated into modern classifications. In order to standardize the terminology used for description, a universal use of the terms “type” and “subtype” was adopted instead of old-fashioned terminology like “entity” and “variant”. Moreover, tumors that were classified independently on the basis of their high grade anaplastic features were reclassified into one tumor type with different biology.

The reorganization of the structure makes sure there is no redundancy or ambiguity in the diagnostics. The overall effect of the WHO fifth edition classification can be seen as a considerable step forward for the implementation of precision oncology in neuro-oncology, with an establishment of the diagnosis model based on tumor biology and its therapeutic stratification.

1.2. Prognostic factors

Prognostic factors are considered key components in the process of making clinical decisions and treatment plans for patients who suffer from brain tumors. It is very important to know about these clinical factors since this allows for the appropriate stratification of patients depending on their specific course of disease [16]. In addition, an adequate approach to predict survival and response to therapy is necessary [17]. A number of clinical and molecular markers have shown great effectiveness in predicting patient survival prospectively in cases of glioblastoma. Some of the essential markers include the age of the patient, isocitrate dehydrogenase mutations, remaining tumor volume, degree of resection, and promoter methylation of O6-methylguanine-DNA methyltransferase (MGMT) [7]. Interactions of these variables have made great progress in improving prognosis in modern neuro-oncological practice. This is exemplified by a cohort study that looked at 186 patients who were diagnosed with newly discovered glioblastomas and subjected to resection surgery and adjuvant chemoradiation. This study has shown that predictors of survival vary essentially in terms of the presence or absence of methylation in the MGMT promoter region.

More precisely, while MGMT promoter methylation is an important predictor of survival in patients treated by alkylating chemotherapy, the volume of surgery becomes the key predictor of survival in patients with unmethylated MGMT. Moreover, there was evidence through which the level of resection became an independent predictor of extended overall survival only in MGMT-unmethylated patients. Thus, there is need to focus on optimal surgical cytoreduction of this particular population because they have shown such potential. The point is that the outcome of preoperative molecular profiling should affect both the choice of treatment and the performance of surgery. Taken together, all of these studies show that the prognosis of glioblastoma is inherently multidimensional. This being said, due to the complex biology of the disease, incorporation of relevant molecular biomarkers along with surgical and clinical-pathological factors is essential in designing an effective

Table 3
Prognostic factors and its clinical implications.

Prognostic Factor	Description	Importance	Clinical Implications/ Considerations
Extent of Resection (EOR) [7]	Degree to which the tumor is surgically removed, categorized as gross total resection, subtotal resection, or biopsy A measure of functional status ranging from 0 to	Strong predictor of survival and recurrence	Complete resection associated with better outcomes but may not always be feasible due to tumor location or size
Karnofsky Performance Score (KPS) [18]	100, where higher scores indicate better functional status Presence of specific genetic alterations in the tumor, such as IDH mutations or MGMT promoter methylation	Key indicator of overall health and treatment response Provides insight into tumor biology and prognosis	Higher KPS associated with better prognosis and increased treatment options Molecular profiling guides treatment decisions and may predict response to targeted therapies
Genetic Mutations [19]	Classification of the tumor based on its microscopic appearance, such as glioblastoma, meningioma, or astrocytoma Assessment of the tumor's aggressiveness based on cellular characteristics and growth patterns	Determines treatment approach and prognosis Higher grade tumors associated with poorer prognosis	Certain histological subtypes associated with better or worse outcomes, influencing treatment strategies Grade influences treatment decisions and may indicate the likelihood of recurrence or metastasis
Tumor Histology [20]	Methylation status of the MGMT promoter region, which affects response to alkylating agents such as temozolomide Sub classification of tumors based on molecular markers, such as IDH	Predicts response to chemotherapy and overall survival Helps refine prognosis and guide treatment selection	MGMT promoter methylation linked to better outcomes and increased sensitivity to certain treatments Different molecular subtypes may have distinct clinical behaviors and responses to therapy
Tumor Grade [21]	mutation status or 1p/19q co-deletion		
MGMT Promoter Methylation [5,22]			
Molecular Subtypes			

treatment plan. The following Table 3 summarizes the main factors affecting prognosis of glioblastomas. The main technical approaches for biomarker discovery and validation are described in Table 4. Every approach is discussed in terms of its utility in clinical settings and strengths and weaknesses.

Optimal safe surgery is a basic cornerstone in the treatment of glioblastoma that seeks to achieve maximum removal of the tumor cells while preserving vital neurological function. Maximum resection has a strong association with better overall survival, especially when carried out along with adjuvant therapy involving chemo radiotherapy. Apart from the factors that are intrinsic to surgery, other clinical factors such as tumor size, baseline Karnofsky performance status, and age are predictive of a lot. The reason for this is because of their intricate relationship with the methylation status of O6-methylguanine-DNA methyltransferase (MGMT). The other important issue concerning molecular stratification of glioblastoma patient samples is an absolutely crucial aspect in the contemporary context of neuro-oncology. Thus, such mutations of the isocitrate dehydrogenase 1 (IDH1) gene serve as a characteristic molecular marker for glioblastomas previously known as secondary cancers with a definite prevalence ranging from 73% to 85%

Table 4
Biomarker discovery and detection techniques.

Technology/ Instruments	Applications	Advantages	Disadvantages
Mass Spectrometry	Proteomics, Metabolomics	High sensitivity and specificity	Expensive equipment, requires expertise
Polymerase Chain Reaction (PCR)	Genetic and infectious disease diagnosis	Rapid amplification of DNA/RNA	Susceptible to contamination, limited multiplexing
Next Generation Sequencing (NGS)	Genomic analysis, cancer profiling	High-throughput, comprehensive analysis	Expensive, data analysis challenges
Microarrays	Gene expression profiling	Simultaneous analysis of thousands of genes	Limited dynamic range, high background noise
Enzyme Linked Immunosorbent Assay (ELISA)	Protein quantification, biomarker validation	High specificity and sensitivity	Requires specific antibodies, limited multiplexing
Flow Cytometry	Cell surface marker analysis	High throughput, single-cell analysis	Limited to surface markers, complex data interpretation
Immunohistochemistry (IHC)	Tissue-based biomarker analysis	Visualizes protein expression in tissues	Subject to variability, requires skilled interpretation
Nuclear Magnetic Resonance (NMR)	Metabolomics	Non-destructive, high resolution analysis	Expensive instrumentation, low sensitivity

[23]. The high prevalence serves as evidence of the peculiar biology and pathology of gliomas containing mutations of isocitrate dehydrogenase 1 (IDH1) genes. Malignancies having IDH1 mutation present with a different metabolic programming, distinct epigenetic patterns, and a much more favorable prognosis in comparison to isocitrate dehydrogenase (IDH) wild type variants.

Thus, the detailed identification of the status of the isocitrate dehydrogenase 1 (IDH1) mutation in glioblastomas has completely revolutionized the modern classification and prediction strategies for disease under the new molecularly oriented WHO classification system. In addition to its practical use in the diagnosis process, the presence of an IDH1 mutation holds important therapeutic significance due to the fact that it plays an important role as a predictor of chemosensitivity and offers a potential molecular target for developing new treatments that can effectively capitalize on unique metabolic and epigenetic susceptibilities. Further research into the molecular biology behind IDH1-related glioblastomas is crucial [21–23].

Baseline clinicoradiographic characteristics remain crucial in the treatment of glioblastoma by offering important parameters for diagnosis, prognostication, and therapy monitoring. The tumor grade, especially its non-invasive assessment through sophisticated neuro-imaging techniques, is still very important in choosing the right therapy. In view of the paucity of prospective studies based on large multicenter clinical trials assessing specific prognostic parameters in glioblastoma, the thorough study of carefully selected retrospective cohorts becomes a necessity. Retrospective studies provide valuable data to unravel the complex web of the prognostic parameters that influence this cancer. For example, the immunophenotyping of Cluster of Differentiation 133 (CD133), a prospective marker of cancer stem cells, was performed in a series of patients with primary intracranial tumors [24]. In this study, immunohistochemistry was performed by using automated methods to measure the amount of CD133 expression in cells based on the presence of methylation in MGMT promoter. The data obtained through the

Kaplan-Meier analysis showed that the presence of Cluster of Differentiation 133 (CD133) antigen expression had no significant effect on prognosis for survival in glioblastomas. In contrast, the methylation of MGMT promoter had high correlation with better therapy response and survival, especially for those patients being treated with temozolomide chemotherapy regimen.

In addition, clinical studies have also emphasized the predictive potential of particular somatic genetic abnormalities, especially MGMT abnormalities, in diagnostic and treatment modalities. These observations suggest that although exploring new biomarkers like Cluster of Differentiation 133 (CD133) is critical for understanding the fundamental biology of the tumor, the application of well-recognized molecular biomarkers in terms of their ability to predict patient prognosis and therapeutic options would be far more effective. In summary, the combination of clinicopathologic data, imaging features, and molecular biomarkers represents the basic structure necessary for implementing personalized management strategies for glioblastoma patients.

There is a strong association between the expression levels of certain HOX stem cell genes such as Homeobox A5 (HOXA5), Homeobox A7 (HOXA7), Homeobox A10 (HOXA10), Homeobox C4 (HOXC4), and Homeobox C6 (HOXC6) and Cluster of Differentiation 133 (CD133) [25]. These studies reveal that the development and progression of glioma occur through a series of biological processes, which are associated with the interaction between CD133+ glioma subpopulations and stem cell signaling pathways. The unique expression pattern of these HOX genes represents an important correlation with overall survival outcomes of patients, which underscores the usefulness of these mRNAs as potential biomarkers for predicting prognosis. Moreover, the expression levels of these Homeobox (HOX) genes have been shown to affect the predictive value of CD133 in these cancers in a reciprocal fashion.

In order to assess differential expression of probe sets between the specified groups of patients, two-sample Welch's *t*-test was carried out, allowing avoidance of type I errors due to differences in variances. The gene-by-gene analysis of probe sets was conducted using a custom-made Java software tool, whereas the obtained data were compiled into tab-delimited sheets.

The list of genes correlated with CD133 was sorted according to increasing *p*-value with emphasis on genes that correlated significantly with CD133 in all three independent groups of patients. Furthermore, the degree of CD133 expression varied significantly depending on tumor grade as assessed histopathologically ($P < 0.001$); this relationship is presented graphically in Fig. 2.

The retrospective study analyzed patients with oncological diseases who were followed up from 2008 to 2011 in the University Hospital of Parma, Italy, with follow-up lasting until 2012 [26]. The method used involved volumetric evaluation of neuroimaging before and after treatment with the help of Voxar 3D software (Ebit AET). The aim of this analysis was to compare the efficiency of radiotherapy with that of

combined radiotherapy and temozolomide therapy. The obtained information gave vital measurements for evaluating the efficacy of multimodal therapeutic protocols for glioblastoma therapy. Simultaneously, experimental evidence proved that the basic stage of treatment of glioblastoma should be primary surgery performed in such a way to obtain maximum cytoreduction [24]. Also, the clinical evidence showed that the development of neuropsychiatric disorders such as anxiety and depression is an important prodromic sign of glioblastoma.

Similarly, a parallel study in which the therapeutic use of an investigational protocol for glioblastoma was applied clinically produced positive outcomes in that there were better overall survival rates and, in selected groups, reduced need for immediate surgery [27]. Also, the therapeutic potential of different particle treatments such as proton beam, carbon ion, and boron neutron capture therapy (BNCT) were reviewed [28]. It has been found that the use of multimodal particles has triggered the activation of several biochemical pathways, thereby providing more effective treatment than photon-based external beam radiation therapy methods. Therefore, current research efforts emphasize the use of multimodal magnetic resonance imaging, metabolic imaging, and functional neuroimaging in order to delineate the target volumes for radiation therapy. The steps in the diagnosis of adult diffuse glioma are shown in Fig. 3.

The data obtained on the basis of the current molecular analysis show that the detection of isocitrate dehydrogenase (IDH) mutations in cerebrospinal fluid can serve as a feasible approach for the non-invasive monitoring of gliomas' development [29]. Thus, it may be concluded that the implementation of molecular diagnostic tools is required for the development of individual treatment strategy.

The biochemical processes associated with the link between isocitrate dehydrogenase (IDH) mutations and glioma formation is one aspect that has been heavily evaluated in recent times. The production of an oncometabolite known as 2-hydroxyglutarate (2-HG), which is a direct product of IDH mutations, leads to the change in the biological behavior of the glioma stem cells and leads to rapid proliferation of the cells under certain microenvironmental settings. IDH mutations also lead to upregulation of the expression of Vascular Endothelial Growth Factor (VEGF). Structurally, there is a definite missense mutation of the isocitrate dehydrogenase 1 (IDH1) enzyme, which is characterized by the replacement of arginine with histidine at amino acid 132.

This unusual clinical scenario of co-occurrence of mutations on both isocitrate dehydrogenase 1 (IDH1) and isocitrate dehydrogenase 2 (IDH2) genes highlights the strict exclusivity of these mutations and their unique pathogenic functions in glioma development. Together, such biological observations elucidate the complicated process of diffuse glioma development and pinpoint potential therapeutic targets. Indeed, in this regard, the epigenetic inhibition of MGMT by promoter methylation plays an equally important role as a predictive and prognostic factor along with IDH mutation status. Further molecular markers included in the central nervous system taxonomy that have been noted in addition to the aforementioned include homozygous deletions of the Cyclin-Dependent Kinase Inhibitor 2 A/2B (CDKN2A/CDKN2B) locus as a stand-alone grade-4 marker for IDH mutant astrocytomas. On the other hand, some of the specific markers for isocitrate dehydrogenase (IDH) wildtype diffuse astrocytomas include 1p/19q codeletions, somatic mutation of Telomerase Reverse Transcriptase (TERT) promoter region, amplification or mutation of the Epidermal Growth Factor Receptor (EGFR) gene and copy number alteration with +7/−10.

Also, comprehensive genomic analysis has established the importance of Telomerase Reverse Transcriptase (TERT) promoter mutations in terms of both diagnosis and prognosis for patients with diffuse gliomas, highlighting their mandatory use in multi-parametric molecular classification algorithms [19]. The integration of these molecular markers in clinical decision-making procedures directly improves the diagnostic accuracy and enables more precise patient risk stratification and proper treatment choice.

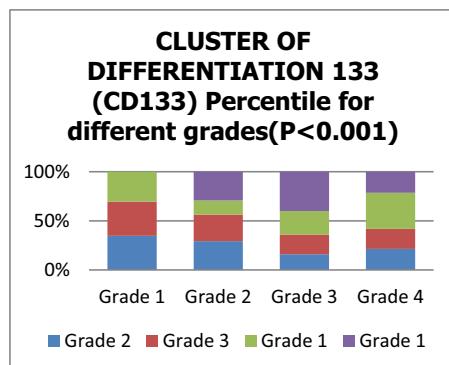


Fig. 2. Cluster Of differentiation 133 (CD133) percentile for different grades($P < 0.001$).

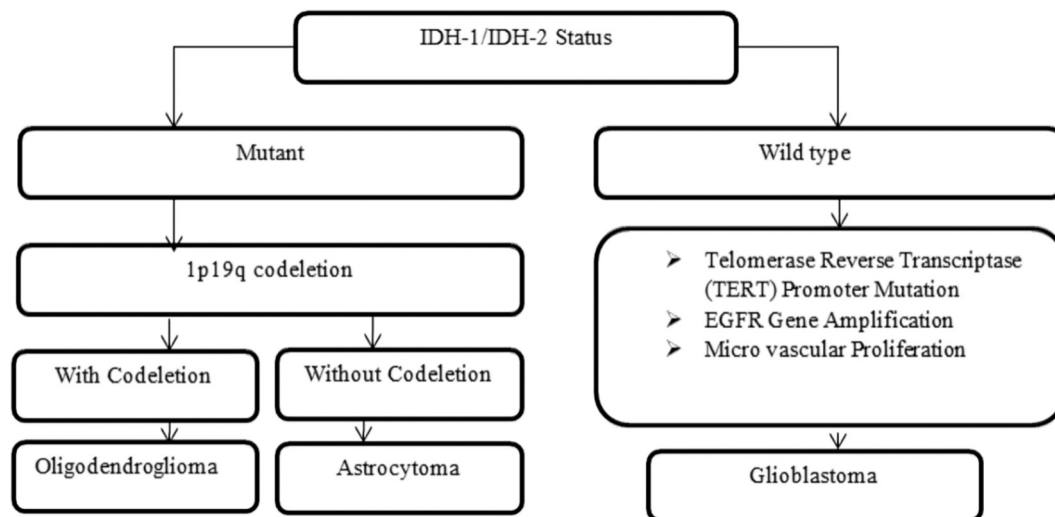


Fig. 3. Diagnostic tree for diffuse glioma of the adult type.

2. Literature survey

Currently, invasive tissue biopsy is an important part of the procedure that helps to determine the exact genetic and molecular signature of almost all brain cancers. However, there is recent information proving that molecular characterizations of tumors can be made without the need for invasive procedures using neuroimaging techniques. This new paradigm relies on using characteristics of molecular imaging to gain significant knowledge about the nature of tumors that help in making personalized treatment decisions in the field of neuro-oncology. In addition, neuroimaging still plays an important role in the clinic in determining sub-cohorts of patients that will benefit from new drugs [30]. Numerous research projects are dedicated to the identification of unique radiological phenotypes associated with certain genetic biomarkers, like Isocitrate Dehydrogenase (IDH) mutation and 1p/19q codeletion, due to their tremendous diagnostic and therapeutic potential. Thus, advanced medical imaging equipment has become an integral tool for non-invasive molecular tumor typing, at the same time enabling optimal localization for stereotactic biopsies in different types of tumors.

Notably, the fast-paced development of artificial intelligence, high-throughput radiomics, and radiogenomics opens promising perspectives for increasing the applicability of state-of-the-art neuroimaging in clinical neuro-oncology. On the other hand, traditional chemotherapy is based on cell cycle disruption or inhibition of blood vessels formation – nonspecific mechanism of action which affects both cancer and healthy parenchymal cells equally [31]. In order to counter these shortcomings, the latest cancer therapies that treat brain cancer include the use of molecularly targeted inhibitors, electric fields for tumor treatment, and tumor-specific vaccine systems. For example, the usual therapy protocol for recurrent glioblastoma often uses the monoclonal antibody, bevacizumab, which selectively targets Vascular Endothelial Growth Factor (VEGF) to inhibit tumor angiogenesis [32]. This anti-angiogenic approach marks a significant technical achievement in the field of neuro-oncology through inhibition of angiogenesis within the immediate tumor microenvironment. These genetic changes cause the primary origin of cancer, progression of the tumor, and subsequent sensitivity to treatment. In addition, different groups of patients have different genetic characteristics which classify certain types of cancers. The genetic changes include the fusion of Fibroblast Growth Factor Receptor 3 (FGFR3) and Transforming Acidic Coiled-Coil Containing Protein 3 (TACC3) genes, and mutation in TERT promoters, which is a common characteristic in adult-type diffuse gliomas. On the other hand, the pediatric-like diffuse gliomas often demonstrate unique mutations at the H3 K27 site and particular EGFR mutations. This difference between the

biological aspects necessitates the need for establishing the molecular framework of brain tumors in order to implement individualized treatment regimes for effective management of the disease.

The investigation of complex regulatory pathways that regulate MGMT expression in brain tumors has shown that transcription factors such as Nuclear Factor- κ B (NF- κ B) and Specificity Protein-1 (SP1) directly stimulate the MGMT promoters [33]. This knowledge is important for understanding the epigenetic regulation of MGMT and for the determination of its biological function regarding chemotherapeutic resistance. Simultaneously, a high rate of mutations and gene amplifications of Epidermal Growth Factor Receptor (EGFR) is observed in glioblastoma cell lines, which is an indication that such genetic changes have a high prognostic significance [34]. Considering that proliferation and survival of cells are highly regulated by internal growth factors, the activation of the aberrant Epidermal Growth Factor Receptor (EGFR) signal transduction pathway leads to important biochemical events.

Evidence gathered shows that Isocitrate Dehydrogenase (IDH) mutations appear in just a few patients with primary glioblastoma, while they present in around 73% to 85% of patients with secondary glioblastoma [35]. The difference between the two types of the disease is a reflection of their biological heterogeneity and points to the importance of molecular stratification when designing treatment. On the other hand, the usefulness of glial fibrillary acidic protein (GFAP) as a systemic biomarker that can be used for diagnosis is hampered by the fact that the protein's expression is highly inconsistent among different heterogeneous regions of tumors, making it impossible to shed consistently in the peripheral blood stream [36]. The diagnostic limitation mentioned above, which can be described as sensitivity deficit, requires identifying additional biological markers in circulation. Taking into consideration the aforementioned need, assessing Telomerase Reverse Transcriptase (TERT) promoter mutations using liquid biopsies is a promising diagnostic approach due to the frequency of these mutations [19,37]. Additionally, isolation methods that concentrate on circulating tumor cells (CTCs) arising from either primary or secondary brain cancers offer an objective method of tumor monitoring, since these cells disassociate from the main tumor mass and can be found in accessible bodily fluids like peripheral blood, cerebrospinal fluid, or urine [38].

The role of CTCs as independent predictors in various cancers has been conclusively demonstrated in a broad range of cancers, such as malignant melanoma, non-small cell lung carcinoma, and pheochromocytoma [39]. The presence of circulating cell-free tumor DNA (ctDNA) has been reported in peripheral blood storage sites in relation to brain tumors, particularly oligoastrocytoma and astrocytomas [34]. Interestingly enough, the levels of ctDNA are still quite low in terms of

absolute concentration in the blood of patients diagnosed with glioblastoma when compared to systemic solid tumors, but at the same time, they still possess a statistically significant positive index allowing for distinguishing this brain cancer from any other type of tumor.

For improved sensitivity in diagnosis, modern-day schemes in neuro-oncology have come up with a two-stage process of classification involving the combination of classical histopathology and tumor genotyping [40]. This multi-parametric strategy has become an operational necessity in overcoming a major problem that existed in diagnosis in that similarities in morphological features of brain and spinal cord lesions led to a misdiagnosis rate of about 10%. Simultaneously, numerous research programs have been devoted to the molecular classification of pediatric brain tumors. As per the newly revised taxonomy of childhood medulloblastoma, improved molecular classification points to a molecular overlap region between Groups 3 and 4, but the WNT/SHH-non type remains the most common clinical subtype. Moreover, among the Sonic Hedgehog (SHH) subtype, different survival outcomes in patients with wild-type and mutated TP53 have led to the distinction of these two types of cancers into separate categories. On the contrary, cytogenetic risk stratification has revealed different high-risk subgroups among patients categorized as posterior fossa ependymoma group A (PFA), owing to the occurrence of recurrent large-scale chromosomal abnormalities including the deletion in chromosome 6q, 16q, and 10q accompanied by 1q gain [41].

3. Approaches and methods

A systematic review of the literature was conducted using the following databases: PubMed, ScienceDirect, and ProQuest. Studies from 2000 to March 2025 that were peer-reviewed were included in this study. Search strategies used for this analysis consisted of carefully selected keywords such as brain tumor, meningioma, glioblastoma, intracranial tumors, brain cancer, outcome evaluation, treatment outcome, and functional outcome. This search strategy enabled the retrieval of pertinent literature relating to multi-faceted aspects of neuro-oncology, treatment effects and functional assessment of patients. In order to determine the risk factors for mortality, multivariate analysis was conducted using Kaplan-Meier survival curves and log-rank test as well as Cox proportional hazards regression [42].

Meanwhile, a thorough analysis of changing trends in the area of automatic brain tumor segmentation techniques revealed significant progress in methodology recently [43]. The development of this technology is mainly fueled by the presence of large imaging databases that have been heavily labeled, along with the fast pace of development of computational segmentation architectures. In particular, the use of deep neural networks allows for processing large data repositories effectively. The popularly used architectures in this computation field include Monte Carlo (MC) dropout techniques, aleatoric uncertainty quantification, and ensemble uncertainty quantification approaches. Most of these computational models adopt the use of convolutional neural networks (CNNs) or other machine learning algorithms such as Naïve Bayes (NB), Random Forest (RF), and Support Vector Machines (SVM). On the contrary, there is a consensus from the existing body of literature that clustering algorithms are seldom used for automatic tumor segmentation tasks.

3.1. Description of the state and prognostic factors

Clinical studies have revealed that young patients respond better to treatment at the onset than their older counterparts. According to epidemiologic studies, about 25% of people undergoing treatment through cytotoxic chemotherapy live for at least two years after diagnosis; this is a considerable improvement on the mere 10% two-year survival rate in people only undergoing external beam radiotherapy treatment. In terms of structure, glioblastoma tumors occur mainly in the supratentorial region, while cerebellar involvement is very uncommon. The

basic histopathological categorization of primary glioblastoma versus secondary glioblastoma was first developed in 1940 by Hans Joachim Scherer, a neuropathologist from Antwerp, Belgium. The early theoretical classification set the fundamental basis for the current classification of glioblastomas as unique and diverse diseases. Moreover, the statistics gathered worldwide by cancer registries reveal that primary intracranial tumors form a large part of the new cases that occur each year, and are responsible for a significant portion of the mortality rate within the field of oncology.

According to global epidemiological evidence, primary brain tumors are responsible for about 28,000 annual incidences of cancers, leading to an annual mortality rate of over 24,000 cases. Given the importance of brain being the major physiological center for the entire organism, timely diagnosis and intervention can significantly increase the chances of survival. In children, intracranial tumors are the second common type of oncological disorders accounting for about 26% of all childhood cancers. However, secondary brain tumors originating from other parts of the body are called metastatic brain tumors and constitute 40% of all systemic cancers.

Cerebral tumors are classified into gliomas and nongliomas, with the neuroepithelial tumors derived from glia being the commonest primary central nervous system tumors. Glial cells have roles as special parenchyma structural cells, offering mechanical structure and metabolic and electrical insulation for neuronal cell circuits. In these neuroepithelial tumors, oligodendroglial tumors form a common category of neuroepithelial brain tumors that occur in the frontal and temporal lobes. Glioblastoma is a highly malignant type of tumor that results from the abnormal development of glial precursor cells or astrocyte cells. It therefore becomes very necessary to understand the molecular cause, origin, and natural history of these varied tumors in order to formulate their treatment strategies.

3.2. Biomarkers and molecular analysis

Molecular biomarkers have been found to be immensely valuable from the standpoint of diagnostics, prognosis, and therapy in clinical neuro-oncology by understanding the underlying pathology, pharmacodynamics, and pathogenesis [44]. Due to the genetic heterogeneity of diffuse gliomas, finding biomarkers has become extremely necessary to help decide on customized treatment options. The clinical importance of this mandate has become increasingly significant with the advent of immunotherapy, especially after the discovery of the immune checkpoint pathways like the programmed death cell receptor 1 and its ligand (PD-1/PD-L1), which have completely revolutionized the modern-day oncology scenario. Moreover, it is seen from clinical trials that populations of gliomas with deficiencies in major histocompatibility complex class I (MHC-I) are resistant to immunotherapy because they effectively elude immune cells. Therefore, it is very strongly suggested that the development of predictive biomarkers be pursued rigorously for the effectiveness of immunotherapy in glioma patients.

Among various biomarker categories, the use of protein-based markers is prevalent in the identification of whether a primary intracranial tumor exists and what its cellular origin and pathology may be. The polypeptide expression patterns present within the tumor tissue provide crucial markers for diagnosis and prognosis. The truncated form of the epidermal growth factor receptor variant III (EGFRvIII) neoantigen, for instance, is common in glioblastoma tumors [29]. On the other hand, biomarker proteins that serve as additional markers include the Ki-67 cellular proliferation index and the O6-methylguanine-DNA methyltransferase (MGMT). The MGMT, located on the 10q26 chromosome, is an enzyme involved in DNA repair by removing alkyl groups from the O6 position of guanines directly through DNA repair in the cell nucleus [45]. It protects cells from dying and is one of the mechanisms involved in developing resistance against chemotherapy drugs. Consequently, changes in the expression levels of MGMT and methylation status play important roles in predicting the survival rate and treatment

Table 5
Biomarkers and key importances.

Biomarker	Subtypes	Key Parameters	References
Telomerase Reverse Transcriptase (TERT)	Telomerase Reverse Transcriptase (TERT) promoter mutation	Impact on tumor progression, survival rates	[5]
IDH	IDH1, IDH2	Mutation frequency, prognostic value	[18]
H3 K27M	–	Diagnostic significance, association with pediatric tumors	[22]
EGFR	EGFR amplification, EGFR mutation	Therapeutic targeting, prognostic significance	[43]
MGMT	Methylated, Unmethylated	Predictive value for chemotherapy response	[46]
GFAP	–	Diagnostic utility, association with tumor types	[47]
Cluster Of Differentiation 133 (CD133)	–	Stem cell marker, prognostic value	[48]
HOX genes	HOXA5, HOXA7, HOXA10, HOXC4, HOXC6	Relationship with tumor aggressiveness, survival	[49]

efficacy in patients with glioblastoma. Details regarding these known biomarkers and their significance in terms of clinical aspects are provided in Table 5 below.

In addition to protein biomarkers, epigenetic biomarkers are essential tools for providing diagnostic and prognostic information since they regulate the regulation of gene expression without changing the DNA sequence itself. Epigenetic factors include different histone modifications, DNA methylation at specific loci, and non-coding RNAs such as miRNAs; they help classify tumors, guide therapy, and predict patients' survival [50]. Thus, the inclusion of an epigenetic analysis strategy in diagnostic protocols is a very efficient way to practice clinical neuro-oncology.

Moreover, metabolic biomarkers provide additional help to molecular diagnostics by making it possible to accurately assess biochemical changes that take place in the tumor microenvironment. It is possible to use advanced imaging methods, for instance, magnetic resonance spectroscopy (MRS), to detect and measure various metabolic levels, like lactate, creatine, choline, and *N*-acetylaspartate. Measurement of metabolic changes makes it possible to distinguish between cancerous and normal brain parenchyma and non-cancerous diseases.

On the other hand, the detection of circulating biomarkers acts as a very efficient way of studying brain tumors without causing harm to the patient. Detecting the various biomolecules in peripheral blood and cerebrospinal fluid in terms of circulating proteins, cell-free circulating tumor DNA (ctDNA), tumor-specific nucleic acids, exosomes, and circulating tumor cells (CTCs) gives real-time information concerning the development of the tumor [51]. The liquid biopsy systems that use these circulating biomarkers act as very viable ways of diagnosing brain tumors without harming the patient. These circulating biomarker categories and their diagnostic ability in neuro-oncology are summarized in Table 6.

Computational approaches for analyzing molecules have become an important tool in the diagnosis and staging of brain tumors, alongside the conventional approach of molecular biomarkers used in the laboratory. Numerous research efforts are currently underway to analyze different panels of tumor biomarkers, with the aim of maximizing sensitivity during diagnosis and developing better staging techniques for

Table 6
Groups and performance of markers in brain tumors.

Tumor Markers	State of Markers	Category	References
O-6 methylguanine-DNA methyltransferase	Act as a predictive and prognostic biomarker	Molecular Biomarkers	[33]
Tumor Protein53(TP53)	Used as a prognostic biomarker		[36]
Telomerase reverse transcriptase(Telomerase Reverse Transcriptase (TERT))	Used as a prognostic biomarker		[52]
Loss of heterozygosity(LOH)	Act as a Prognostic and Predictive Biomarker		[53]
Glial Fibrillary acidic proteins(GFAP)	Used as a Prognostic Biomarker		[54]
Circulating extracellular vesicles	Used for Predictive and Monitoring response to treatment	–	[55]
Circulating free DNA	Used for a diagnosis and monitoring response to treatment	–	[39,56]

adult brain tumors [57]. In the field of computation, the Genetic Algorithm-U-Net (GA-UNET) has emerged as a highly integrated model, showing better performance metrics than other competing molecular models.

In this computational process, the radiological images are optimized by employing Contrast Limited Adaptive Histogram Equalization (CLAHE) after the automated exclusion of all other body parts apart from the brain through the automated skull-stripping process. The process of optimizing the contrast is made possible by calculating the histograms of local pixel intensities in predefined regions. Phenotypes on brain imaging studies can be heavily affected by somatic genetic changes. For example, mutations in the promoter region of Telomerase Reverse Transcriptase (TERT) are very common among different intracranial tumors, functioning as independent predictive factors allowing us to create more advanced molecular tests [19]. Other important molecular abnormalities affecting the phenotype include MGMT promoter methylation, IDH1 and IDH2 mutations, and deletion of the short arm of chromosome 1 and the long arm of chromosome 19 (1p/19q codeletion). This means that making an accurate diagnosis of particular central nervous system cancers requires proving the presence of these genomic alterations.

Mechanistically, a confirmed 1p/19q codeletion entails the deletion of both chromosome arms of 1p and 19q, being the distinctive marker for oligodendroglioma. For patients with anaplastic oligodendroglioma, the significance of this particular chromosomal aberration cannot be overlooked in terms of treatment options. Whereas certain genetic influences exist on the development of cancers within the central nervous system, detecting low-penetrance polymorphisms in sporadic cases is a difficult task. However, highly penetrant germline mutations cause certain syndromic neurological tumors. The neurofibromatosis 1 condition originates as a result of inherited mutations that lead to inactivation of the Neurofibromatosis type 1 (NF1) tumor suppressor gene, while neurofibromatosis 2 is characterized by germline mutations associated with the Neurofibromatosis type 2 (NF2) gene locus. In addition, the clinical relationship between diffuse gliomas and hereditary conditions is exemplified in the classical case of the Li-Fraumeni syndrome, which is caused by germline mutations in the tumor protein p53 (TP53) gene. Likewise, the increased occurrence of primary brain and spinal cord tumors is associated with Cowden and Gorlin hereditary syndromes.

3.3. Neurological test

Discrete deficiencies of particular neurological faculties give key localized indications regarding the location of the glioblastoma tumor within the brain. This application is possible since a comprehensive neurological examination methodically assesses variables related to vision, deep tendon reflexes, balance, motor coordination, and cognitive function, all of which serve to identify the precise localization of the tumor. Research shows that a primary oncogenesis of the central nervous system secondary to syndrome-related genetic malfunctions is uncommon, with 5% of such cases being due to genetically identifiable germline mutations. Though certain susceptible genes may be passed down to offspring, the great majority of brain tumors appear as sporadic occurrences among patients having no previous familial history of brain cancer.

Physically, recent therapeutic innovation includes FDA approval of the Optune technology in 2015, a non-invasive treatment utilizing electric tumor-treating fields for treating newly-diagnosed glioblastomas [58].

3.4. Biopsy

Histopathologic tissue sampling through neurosurgery can be done quite often since it allows for obtaining histological samples for examination under the microscope and thus determining the benign or malignant nature of an intracranial mass. Stereotactic brain biopsy can be considered one of the major minimally invasive neurosurgeries used for making a definitive diagnosis of deep-seated or eloquent intracranial masses [59]. In this case, a computer-assisted three-dimensional coordinate system is used to determine the volume of interest within the brain [60].

In comparison to open craniotomy and surgical resection, the use of stereotactic biopsy procedures results in minimal damage to the parenchyma and much faster recovery times after surgery. Nevertheless, accurate patient monitoring is required since there are perioperative complications that may arise including intracranial bleeding, worsening of cerebral edema adjacent to the lesion, local tract infection, and side effects from general anesthesia. In spite of all these complications, stereotactic biopsy procedures are an absolute necessity in order to elucidate the biological and molecular structure of the tumor in question. Neurological deficits arising after such procedures tend to be transient and can be managed using pharmacologic means. It should be noted that adult brain cancers do not necessitate any systemic staging workup due to their lack of tendency to metastasize.

3.5. Partial

Total surgical excision is often impossible due to structural involvement or complex anatomical structures which allow for only partial tumor reduction. Primary brain cancers arise from any anatomical structure of the nervous system. Nevertheless, some brain cancers

arise at deep or crucial sites, making the treatment of such tumors difficult as the lesions arise right next to eloquent brain cortex areas [61].

As a result, the use of quantitative imaging biomarkers in clinical practice has gained significance not only in terms of predicting survival but also in assessing long-term treatment efficacy. High relative cerebral blood volume (CBV) measured using perfusion-weighted brain imaging is highly correlated with poor clinical outcome in patients with glioblastoma; therefore, increased CBV in certain intratumoral sub-volumes is indicative of resistance to conventional treatment protocols [26,29,60,62]. Furthermore, high b-value diffusion-weighted imaging, used for diagnosis, proves to be more efficient than conventional ADC mapping when discriminating between tumor recurrence and treatment-related radiation effects; it demonstrates higher diagnostic accuracy at 89% against 67%. Table 7 outlines some genetic diseases and the respective signal alterations involved in brain tumor development.

Comprehensive genomic profiling has revealed the existence of distinct somatic alterations in the neurofibromin 1 (NF1), Erb-B2 receptor tyrosine kinase 2 (ERBB2), and phosphoinositide-3-kinase regulatory subunit 1 (PIK3R1) genes in particular molecular subtypes of glioblastoma. Apart from the recurrent mutational events in the PIK3R1 locus, a seminal molecular study conducted through the Cancer Genome Atlas Network has identified the receptor tyrosine kinase/rat sarcoma/phosphoinositide 3-kinase (RTK/RAS/PI3K) signaling pathway as a key molecular pathway disrupted in the development of glioblastomas along with the synchronous mutations in the Erb-B2 receptor tyrosine kinase 2 (ERBB2) and neurofibromin 1 (NF1). Subsequent multi-omic integrated studies have associated the loss of function in the NF1 with the aggressive mesenchymal glioblastoma subtype and the direct oncogenic role of somatic recurrent mutations in the PIK3R1. On the other hand, amplification or point mutations in the ERBB2 gene occur in just a small

Table 8
Prognostic implications of NF1, ERBB2, and PIK3R1 gene mutations.

Gene(s)	Findings	References
NF1	10% to 15% of sporadic high-grade gliomas have somatic NF1 alterations" and that NF1-deficient glioblastoma neurospheres are sensitive to MEK inhibitors + irradiation	[68]
PIK3R1 / PI3K pathway	The analysis of integrating <i>PIK3CA</i> and <i>NF1</i> mutations into GBM prognostication and immunotherapy response: mutated PIK3R1 or PIK3CA occur ~15% of GBM; NF1 mutation or deletion also frequent	[69]
ERBB2 (HER2)	Diverse ERBB2/ERBB3 activating alterations across solid tumors": though not GBM-exclusive, ERBB2 has activating genomic alterations in solid tumors and emphasizes the importance of detecting those even when traditional HER2 overexpression or amplification is low. That suggests ERBB2 could also be relevant in GBM if screened.	[70]

Table 7
Genetic disorders and signaling conditions.

Oncogene	Gene Encoding	Gene Locus	Mutant Protein Function	Syndrome	References
TP53 (tumor suppressor gene)	p53	17p13.1	Cells with damaged DNA are proliferated and survive which contribute to the transformation of a normal cell into the malignant cell	Li-Fraumeni syndrome	[63]
VHL (tumor suppressor gene)	pVHL	3p25	Mutant pVHL leads to HIF overexpression and upregulation of VEGF, platelet-derived growth factor and erythropoietin. pVHL inactivation is associated with the angiogenesis	Von Hippel-Lindau	[64]
TSC1 and TSC2	TSC1 encode hamartin TSC2 encode tuber in	TSC1 on 9q34 TSC2 on 16p13.3	Mutation in TSC1 and TSC2 disrupts the endogenous gene functions in suppressing the downstream mTOR activity through different mechanisms	Tuberous sclerosis	[65]
NF2 (tumor suppressor gene)	Merlin	22q12	Mutant merlin decreases the transcript expressions which inhibit the canonical Wnt and RhoA signaling pathway. This mutation also induces alterations in gene expression	Neurofibromatosis type 2	[66,67]

fraction of glioblastomas and possibly act as a molecular determinant for treatment resistance in these tumors to conventional treatment regimes [68]. The chronological summary of the key findings and implications of the recent molecular studies on the above-listed genomic alterations is presented in Table 8 below.

Genomic analysis at large scale has shown that the incidence of mutations in neurofibromin 1 (NF1), Erb-B2 receptor tyrosine kinase 2 (ERBB2), and phosphoinositide-3-kinase regulatory subunit 1 (PIK3R1) in patients with glioblastoma or high-grade glioma is clinically relevant. Mutations or deletions occur in the NF1 gene in about 10% to 15% of sporadic cases of high-grade gliomas. In addition, deficiency of NF1 increases sensitivity of glioblastoma neurospheres to combination therapy involving MEK inhibition and ionizing radiation. At the same time, about 15% of glioblastomas display somatic mutations and copy number changes in the phosphoinositide-3-kinase regulatory subunit 1 (PIK3R1) gene which tend to be associated with mutations in the PIK3CA gene, the phosphoinositide-3-kinase catalytic subunit alpha. The overlapping genetic changes are part of the composition of specific gene signatures that help predict response to systemic immunotherapy [69]. In spite of the fact that the Erb-B2 receptor tyrosine kinase 2 (ERBB2) genes does not receive much genomic characterization in unselected glioblastoma patients, data generated via genome analysis indicate that point mutations and amplifications in the Erb-B2 receptor tyrosine kinase 2 (ERBB2) gene represent actionable molecular targets [70].

4. Clinical applicability and limitations of liquid biopsy in brain tumors

Liquid biopsy acts as an adjunct but not as a substitute for traditional tissue biopsy since it offers a dependable approach towards detecting tumor recurrence, monitoring response to treatment, and tracking changes in tumor biology [71,72]. Liquid biopsy platforms present themselves in clinical applications where there is non-invasive molecular testing at the point of diagnosis in regions of the central nervous system which cannot be accessed surgically. Evidence suggests that assays targeting circulating cell-free tumor DNA (ctDNA) have proven effective in detecting somatic mutations that help in making definite classifications of tumors using World Health Organization fifth edition criteria, which include EGFR gene amplification, TERT promoter mutation, IDH mutation, and histone H3 K27M mutations [13]. Since cerebrospinal fluid is directly located next to the tumor mass and does not face the limitations associated with the blood-brain barrier, the accuracy of ctDNA extracted from cerebrospinal fluid shows high levels of correlation to that obtained from primary tumor compared to plasma-derived ctDNA for pediatric diffuse midline gliomas and adult glioblastomas [73].

Several operational challenges exist that make the use of liquid biopsy technology difficult for the detection of central nervous system cancers owing to diagnostic sensitivity and specificity issues. First, the presence of blood-brain barrier as well as limited release of larger tumor molecules into peripheral circulation makes the detection of tumor cells using the ctDNA technique very inefficient in the context of primary brain tumors with an efficiency rate varying between 10% and 40%. For this reason, cerebrospinal fluid is preferred to be the major source of samples for testing purposes since the tests performed on such a sample are more sensitive than those performed on peripheral plasma with particular regard to high grade gliomas and diffuse midline gliomas.

In addition to this, while ctDNA sequencing provides extremely high specificity for diagnosis once the exact somatic founder mutation is established, the problem of false positives persists due to the existence of clonal hematopoiesis of indeterminate potential (CHIP), which serves as an artificial hematological factor leading to false-positive signals especially during sequencing of blood banks where common mutations in the genes of the tumor protein p53 (TP53) and Telomerase Reverse Transcriptase (TERT) are analyzed. Therefore, strict adherence to assay

design protocols will become inevitable, which should include simultaneous sequencing of matched donor leukocyte DNA. Due to the preservation of complete cellular structures, CTCs have high biological specificity; however, since the CTCs are extremely rare in the systemic circulation of individuals with intracranial tumors, detection sensitivity and assay reproducibility are highly limited. The technical rarity makes CTC-isolation assays significantly less clinically sophisticated compared to the highly-advanced ctDNA technologies used in modern neuro-oncology [72,74].

Other restrictions include the large difference in technological aspects between different platforms for high-throughput sequencing and bioinformatics analysis, along with spatial intratumoral heterogeneity leading to possible under-representation of certain clonal changes from circulating biomarkers' libraries. Moreover, the application of these markers in clinical neuro-oncology is currently limited by the lack of validated analytical thresholds needed to use this information in clinical decision-making according to absolute concentration of cell-free tumor DNA or circulating tumor cells [13,75].

4.1. State of neurologic function

Functioning of patients is assessed using standardized methods, for example, Karnofsky Performance Status (KPS), which measures the functioning of an individual patient depending on his ability to carry out the activities of daily living. This clinical scoring system measures the functioning of a patient on a continuum that ranges from 100, where there is complete physiological function without any illness, to 10, which indicates a moribund state where there is rapid clinical deterioration of the patient. The higher the Karnofsky Performance Status (KPS) score, the better the independence and the higher the chances of survival.

Significant correlations have been found between the results of initial functional assessment and performance measures over time, which indicate favorable results in 78.7% of patients when using the special Nano Scale test, while only 63.8% in case of using the conventional Karnofsky Performance Status (KPS) parameters [18]. Both primary and secondary brain tumors occur in all population groups, yet there are two peaks of occurrence in children and old people. In addition, the general survival is significantly correlated with the KPS score, the total number of cerebral metastases, and the condition of the primary tumor site and the extracranial spread of the disease. This set of clinical factors is analyzed in the literature via univariate and multivariate statistics. In modern oncological care practice, an initial KPS score of 70 or above or an equivalent WHO performance status of 0 to 2 represents a standard mandatory requirement for therapeutic treatment and participation in clinical trials. Additionally, multivariate regression analysis reveals that both older age and specialization within an institution have a significant effect on baseline functional performance scores and trial patient recruitment.

4.2. Assessments: histology

The therapeutic approach towards a vast number of central nervous system tumors remains inadequate, leading to substantial neurological disability and severely reduced life expectancy. Numerous research efforts are being undertaken to develop improved therapeutic strategies in adults and children patients, which are motivated largely by the considerable cellular heterogeneity seen in intratumoral proliferation rates, as well as presence or absence of focal necrosis. In the absence of molecular information, a lesion showing overlapping features is often diagnosed simply as oligoastrocytoma, not otherwise specified (NOS), precluding precise differentiation from true oligodendroglial or astrocytic tumor types.

In terms of statistical considerations, age still stands out as a critical determinant of survival outcomes. The survival rate of pediatric patients below 15 years of age stands at 75% within a five-year period. On the

Table 9

Comparative overview of WHO 2016 vs. WHO 2021 (CNS5) CNS tumor classification.

Aspect	WHO 2016 Classification	WHO 2021 CNS5 Classification	References
Diagnostic approach	Histology and molecular findings often reported separately	Integrated diagnosis combining histology, molecular data, and clinical context into a single diagnostic entity	[13]
Role of molecular markers	Limited to selected tumors (e.g., IDH mutation, 1p/19q codeletion); optional in many cases	Mandatory molecular criteria for defining most major tumor entities	[13]
Interobserver variability	High, especially for diffuse gliomas and grading	Reduced variability due to objective molecular criteria and integrated reporting	[13]
Overall framework	Transitional histomolecular classification; histopathology remained central	Fully integrated, molecularly driven classification with genomics at the core	[13]
Grading notation	Roman numerals (Grade I–IV)	Arabic numerals (Grade 1–4), assigned within tumor types	[13,76]
Handling of ambiguous cases	Frequent use of “NOS” (Not Otherwise Specified) due to missing molecular data	Introduction of “NEC” (Not Elsewhere Classified) and reduced reliance on NOS through defined molecular requirements	[77]
Tumor grading system	Grading largely based on histopathological features (mitoses, necrosis, micro vascular proliferation)	Molecularly informed grading, where specific genetic alterations can define high-grade disease even without histological hallmarks	[13,77]
Definition of tumor entities	Many entities defined morphologically with molecular modifiers	Molecularly defined tumor entities based on recurrent genomic alterations (e.g., H3 K27, MAPK pathway)	[13,77]
Adult vs. pediatric gliomas	Grouped together with age modifiers	Explicit separation into adult-type and pediatric-type diffuse gliomas based on distinct molecular biology	[13,78]
Alignment with precision medicine	Partial	Strong alignment; supports targeted therapies and molecular trial stratification	[13,78]
Obsolete histological categories	Retained several legacy categories based on morphology	Removal or consolidation of obsolete or redundant histological entities lacking molecular distinction	[79]
Clinical relevance	Limited prognostic and therapeutic stratification	Improved prognostic accuracy and treatment stratification, aligning classification with precision oncology	[13,79]

other hand, for adolescent and young adults between 15 and 39 years, their five-year relative survival rate is at around 72%. However, for adults above 40 years, the five-year relative survival rate declines sharply to 21%. A structural matrix highlighting the key diagnostic changes that differentiate the 2016 WHO classification criteria from the current 2021 WHO CNS5 version is summarized in Table 9.

The WHO's 2021 CNS taxonomy reformulated past diagnostic systems through the creation of new classes of diseases characterized by unique molecular profiles and pathological criteria. Within this modern classification system, gliomas with somatic mutations within the Isocitrate Dehydrogenase alleles belong to the one group of “astrocytoma, IDH mutant,” encompassing all WHO grade 2, 3, and 4 central nervous system lesions. Most important, particular genomic changes can be used as individual criteria in this grading system. For instance, somatic homozygous deletion of the CDKN2A/B gene can be considered an individual predictor of an aggressive biological course, and the definitive grade 4 central nervous system WHO classification is required regardless of no microvascular proliferation and necrosis presence. Integration of the particular molecular criteria into standard tumor grading procedures provides the highest level of diagnosing reproducibility and gives a strong base for predicting prognosis.

4.3. Circulating tumor cells

The independent prediction power of circulating tumor cells (CTCs) has been proved for various groups of oncologic disorders such as malignant melanoma, lung carcinoma, and pheochromocytoma [80]. In the context of this technology, CTCs were isolated through magnetic cell separation using 5 mL of peripheral whole blood while EVs were separated simultaneously by membrane affinity isolation from 4 mL of plasma.

Target gene expression analysis was done through reverse transcription, target preamplification and multi-marker qPCR with 18 specific target genes [81]. Unique transcriptional signatures for Aurora Kinase A (AURKA), Poly ADP-ribose polymerase 1 (PARP1), and SRC Proto-oncogene were observed in the selected extracellular vesicle isolates. However, the increased transcriptional activity of genes in the phosphoinositide 3-kinase (PI3K) signaling pathway was evident in the circulating tumor cells. The molecular signature differences highlight the value of using both CTC and EV analysis for comprehensive cancer diagnosis.

4.4. Circulating free DNA

Cell-free tumor DNA (ctDNA) represents fragmented genetic material that has been released into the systemic bloodstream by tumor cells experiencing apoptotic or necrotic cell death. The extraction and analysis of such circulating DNA serve as an invasive molecular tool for the purpose of conducting disease profiling. Several studies have reported the existence of validated ctDNA fragments in the bloodstream of selected patients suffering from brain tumors like oligoastrocytoma and astrocytoma [82]. However, the absolute amount of circulating ctDNA in the blood stream of patients suffering from glioblastoma multiforme is considerably less when compared to the amount found in other solid tumors of extracranial origin. This decrease in circulating genomic DNA results from the anatomical isolation due to blood brain barrier along with other biological features of primary intracranial tumors.

4.5. Circulating free miRNA

MicroRNAs (miRNAs) are vital molecules necessary for sustaining homeostasis within cells and regulating intercellular communication pathways during normal and neoplastic processes [82]. Within the past ten years, various clinical studies have explored the application of miRNAs in serum samples as circulating biomarkers for glioblastoma. In particular, through comprehensive profiling, researchers discovered a

set of differentially expressed miRNAs, which included 115 miRNAs overexpressed and 24 miRNAs under expressed in the serum of individuals suffering from glioblastoma [56].

On the other hand, other molecular investigations have found a considerable decline in the serum levels of miRNA-125b in cases of glioblastoma [55]. All these sets of data indicate that miRNAs can serve as highly promising non-invasive markers for the diagnosis and monitoring of glioblastoma.

4.6. Circulating extracellular vesicles

Cancer-derived extracellular vesicles that are released from cancer cell clusters actively get taken up by adjacent stromal cells located within the corresponding microenvironment, which causes changes in function, directly enabling tumor growth and vascular formation [83]. Higher levels of EVs in the bloodstream of patients suffering from glioblastoma have been identified, and they are higher compared to those of healthy individuals; besides, the increased levels of these vesicles have been shown to be associated with subclinical tumor recurrence after surgery [84]. As a result, extracellular vesicles derived from the tumor are highly effective as a circulating biomarker for monitoring the disease.

Different techniques are often used for the isolation and characterization of the mentioned population of vesicles. The commonly used techniques for the study of EVs include direct visualization using transmission electron microscopy (TEM), estimation of the size and concentration using nanoparticle tracking analysis (NTA), and identification of surface proteins, like intercellular adhesion molecule-1 (ICAM-1) or integrin heterodimers using flow cytometry or quantitative Western blotting [62].

4.7. Circulating proteins

Peripheral blood protein biomarkers that circulate in the bloodstream could be used as a possible tool for assessing the effectiveness of therapy among patients suffering from tumors of the central nervous system [85]. However, certain proteins that have a neurological origin, such as S100 calcium-binding protein B (S100B) and brain-derived neurotrophic factor (BDNF), can be detected in the bloodstream of healthy controls. Therefore, the natural presence of these proteins in the bloodstream reduces their diagnostic sensitivity, thus making them clinically irrelevant tools to detect brain diseases.

5. Conclusion

The complementary somatic genetic changes that underlie glioblastoma pathogenesis include recurring mutations or deletions of the Neurofibromin 1 (NF1), Erb-B2 Receptor Tyrosine Kinase 2 (ERBB2), and Phosphoinositide-3-Kinase Regulatory Subunit 1 (PIK3R1) genomic regions. Such changes are always identified in different molecular classes of glioblastomas and other high-grade diffuse gliomas. At the same time, considerable research interest has been paid to comparing the relative clinical significance of MGMT promoter methylation status and Isocitrate Dehydrogenase (IDH) mutation pattern in regard to their ability to serve as definitive predictors of subsequent therapy results and survival patterns [86]. Additionally, it has been established empirically that early primary tumor removal offers better clinical results compared to the old conservative approach of careful observations used for low-grade diffuse gliomas due to decreased chances of malignant transformation and possibility of further molecular classification [87]. Secondary glioblastomas that carry Isocitrate Dehydrogenase (IDH) mutations develop from lower-grade tumors such as anaplastic astrocytomas and have characteristic Isocitrate Dehydrogenase 1 (IDH1) somatic mutations along with CpG island methylator phenotype (G-CIMP) which makes them biologically different from primary glioblastomas. The mutant IDH produces an oncometabolite called 2-

hydroxyglutamate (2-HG), that blocks the function of demethylases, causing a unique form of hypermethylation of the genome referred to as CpG island methylator phenotype (G-CIMP). They develop through completely different oncogenetic mechanisms and generally have much better prognosis. In terms of the comorbidity of conditions related to the disease in question, current approaches within the field of neuro-oncology have discovered a multitude of different mechanisms that interact to cause epilepsy associated with tumors – namely macroscopic mass effect, hyperexcitability of brain regions next to the tumor margin, changes within the microenvironment in the extracellular matrix, and hyper-stimulation of the glutamatergic *N*-methyl-D-aspartate (NMDA) receptors through metabolic pathways. On the cellular level, glioblastoma cells tend to be especially susceptible to ferroptosis, which is a special type of programmed non-apoptotic cell death mediated by the iron ions and caused by severe lipid peroxidation and altered amino acid transport kinetics. As a result, drug-based induction of cellular ferroptosis becomes an extremely promising area of study. However, although there has been considerable progress on the laboratory level, integration of circulatory and molecular biomarkers into the field of neuro-oncology still needs to evolve because the effectiveness of the former depends on the correct identification of the malignant condition in question.

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P. Balaji: Writing – review & editing. **D. Devasena:** Formal analysis, Supervision.

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Data availability

Data will be made available on request.

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